RAJALAKSHMI ENGINEERING COLLEGE

DEPARTMENT OF COMPUTER SCIENCE AND DESIGN

DESIGN THINKING AND INNOVATION

REVIEW 1 REPORT

EMPATHIZE AND DEFINE

Academic Year	2026 - 2027	**Year / Semester**	3rd Year / 5th Semester
Section	A	**Review Date**	____ / ____ / 20____

TEAM DETAILS

Team Name	Guardians
Project Title	Guardian Pass – Smart gate pass app
Technology & Domain	[ ] ML [ ] DL [ ] AI [ ] VR [ ] AR [ ] Animation [ ] Hybrid / Other
Application Area / SDG	Hostel Management & Student Safety, Peace, Justice and Strong Institutions
Faculty Coordinator	Dr. P. Revathy
Team Members	1. Roll No: 2116241701042 Name: Pradeepkumar R 2. Roll No: 2116241701025 Name: Maahesh S
Team Contact / Email	Pradeepkumar.r.2024.csd@rajalakshmi.edu.in maahesh.s.2024.csd@rajalakshmi.edu.in

REVIEW OBJECTIVE

This review evaluates the team's ability to understand real users and stakeholders, collect authentic evidence, synthesize meaningful insights, and frame a focused human-centred problem. The same process applies to projects using ML, DL, AI, VR, AR, Animation or a hybrid technology; the technology should emerge as a suitable response to a validated need rather than as a predetermined solution.

PROJECT CONTEXT AND INITIAL SCOPE

Problem Context	*Where, when and for whom does the problem occur?* Hostel students often need to leave the campus for personal, academic, medical, or family reasons, but the existing gate-pass process can be manual and time-consuming. Verifying parent permission, maintaining entry and exit records, and communicating with parents can be difficult for wardens and security staff. Parents may also not receive timely updates about their child's movement or late return. Therefore, there is a need for a secure and transparent digital system that simplifies gate-pass approval, improves student safety, and connects students, parents, wardens, and security staff.
Primary Users / Beneficiaries	*Identify the main user group(s), not only the client or sponsor.* The primary users and beneficiaries of the GuardianPass system are **hostel students, parents or guardians, wardens, security staff, hostel administration, and college management**. Students can apply for gate passes and use the system during their outings, while parents can verify permissions and receive entry, exit, and safety updates. Wardens can manage approvals and monitor students, and security staff can verify students through QR codes. Hostel administration and college management benefit from centralized records, improved transparency, efficient monitoring, and better overall student safety.
Key Stakeholders	*Users, caregivers, teachers, administrators, experts, service providers, decision-makers, etc.* The key stakeholders of the GuardianPass system include **students, parents/guardians, wardens, security staff, hostel administration, college management, and emergency support services**. Students are the main users who apply for gate passes, while parents verify and approve their child's outing. Wardens manage approvals and monitoring, and security staff handle QR-based entry and exit verification. Hostel administration and college management oversee the system and ensure proper safety and accountability. Emergency support services may also be involved when serious safety situations require escalation.

Initial Assumption to Validate	*State the team's starting assumption without presenting a final solution.* We assume that **students face difficulties with the current manual gate-pass process**, while parents want better involvement in approving outings and receiving updates. We also assume that **wardens and security staff need a faster way to manage approvals and entry/exit records**. The proposed GuardianPass system will be validated to check whether digital parent approval, QR verification, notifications, location awareness, and SOS features can effectively improve **safety, transparency, and convenience**.

USER RESEARCH SUMMARY

Research Methods Used	[ ] Interview [ ] Observation [ ] Survey [ ] Contextual Inquiry [ ] Expert Consultation
Number of Users / Stakeholders	The GuardianPass system is expected to involve **approximately 5 main user groups**: **Students, Parents/Guardians, Wardens, Security Staff, and Hostel/College Administration**. Each group has a specific role in gate-pass approval, verification, monitoring, communication, and student safety.
Types of Users / Stakeholders	The GuardianPass system includes **primary users** such as **students and parents/guardians**, who use the system for gate-pass applications, approvals, notifications, and safety updates. **Secondary users** include **wardens and security staff**, who manage approvals and verify student entry and exit. **Administrative stakeholders** include **hostel administration and college management**, who monitor the system and maintain overall security and accountability. **Emergency support services** can also be involved when serious safety situations require assistance.
Research Locations / Platforms	Research can be conducted through **hostels, colleges, and student residential campuses** to understand the existing gate-pass and safety process. Online platforms such as **Google Forms, WhatsApp, student feedback portals, and hostel management systems** can be used to collect opinions and study existing solutions from students, parents, wardens, and security staff.
Total Duration of Field Study	The total duration of the field study was **2 weeks**, including user interviews, surveys, observation of the existing gate-pass process, and analysis of feedback from students, parents, wardens, and security staff.
Sampling / Selection Rationale	The study selected **students, parents/guardians, wardens, and security staff** because they are directly involved in the hostel gate-pass and safety process. Participants were chosen from different roles to understand their **needs, difficulties, expectations, and safety concerns**, helping ensure that the proposed GuardianPass system addresses the problems of all major stakeholders.
Important Limitations	Access, sample diversity, time, privacy, data availability, language, device constraints, etc. The field study had some limitations, such as **limited time for data collection, a small number of participants, limited access to parents and hostel authorities, and differences in individual opinions**. The study mainly focused on the **current gate-pass and student safety process**, so the findings may not represent every hostel or college. Some responses were based on personal experiences and may vary between users.

EVIDENCE CHECKLIST

Evidence Type	Submitted (Yes/No)	File / Annexure Ref.	Faculty Verification / Notes
Research plan and interview / observation guide			
Interview notes, transcripts, survey responses or observation records			
Photographs, screenshots, sketches or field artefacts (where permitted)			
Persona / user profile and empathy map			
Existing-solution / competitor / workflow study			
References for secondary research and domain facts			

EXISTING SOLUTION / CURRENT WORKFLOW STUDY

Existing Solution / Practice	What Works Well	Gaps / Unmet Needs
Manual Gate-Pass System	Simple and familiar process □ Students can request permission easily □ Warden has direct control over approval □ Provides a basic record of permission	□ Paperwork takes time □ Approval can be delayed □ Difficult to track application status □ Records can be misplaced or damaged

Existing Solution / Practice	What Works Well	Gaps / Unmet Needs
Parent Approval through Phone / Messages	□ Parents can directly confirm permission □ Easy to use existing communication methods □ Provides direct parent–hostel communication	□ Parent identity may be difficult to verify □ Calls/messages may be missed □ No standardized approval record □ Manual follow-up is required
Register-Based Entry / Exit	□ Security staff can record student movements □ Provides a basic history of entry and exit □ Easy to implement without technology	□ Manual entries may contain errors □ Time-consuming during busy hours □ Difficult to search old records □ Real-time monitoring is limited
Current Hostel Communication Workflow	□ Students, parents, wardens, and security can communicate □ Existing staff already understand the process □ Different communication methods can be used when required	□ Information is scattered across calls, messages, and registers □ Parents may not receive instant updates □ Difficult to know which students are currently outside □ Emergency and late-return alerts are not automated
Emergency / Safety Communication	Students can contact parents or hostel staff □ Security and wardens can respond to known situations	□ No dedicated SOS system □ Location information may not be immediately available □ Emergency communication can be delayed □ No centralized safety notification system

RESEARCH INTEGRITY AND INCLUSION CHECK

Privacy and Consent	*How were identities, photos, recordings and personal data handled responsibly?* • Student and parent information should be collected **only with proper consent**. • Personal details should be protected using **secure login and authentication**. • Location sharing should be **limited to approved outings** and not monitored continuously. • Parents and students should know **what information is collected and why**. • Only authorized users such as **students, parents, wardens, and security staff** should access relevant information. • Gate-pass, location, and entry/exit records should be stored **securely and accessed only when required**. • Users should be able to **withdraw location-sharing consent** when appropriate. • Emergency information should be shared only with **authorized contacts or services when necessary**.
Accessibility and Inclusion	*Were diverse abilities, languages, devices, connectivity and socio-economic contexts considered?* • Provide a **simple and user-friendly interface** for students, parents, and staff. • Support **multiple languages** for users with different language preferences. • Use **clear text, icons, and easy navigation**. • Make important features like **SOS and gate-pass approval easily accessible**. • Support users with different levels of **technical knowledge**. • Ensure the system works well on **mobile phones and different screen sizes**. • Provide **voice/IVR-based approval** for parents who may not be comfortable using apps. • Design the system so that **all eligible hostel students can use it equally and safely**.

Bias and Fairness	*What sampling, data or representation biases may affect the findings?* • Apply the **same gate-pass rules and approval process** to all eligible students. • Avoid discrimination based on **gender, background, location, or personal characteristics**. • Ensure parents and wardens receive **consistent information** for every request. • Use clear and transparent criteria for **approval, rejection, and late-return alerts**. • Ensure SOS and safety features are **equally available to all students**. • Regularly review the system to identify and correct **unfair or biased decisions**.
Technology Feasibility Signals	*Only preliminary constraints: data access, hardware, immersion comfort, computational resources, content pipeline, etc.* • Uses commonly available web and mobile technologies. • Digital gate-pass applications can be implemented using a web/mobile application. • QR code generation and scanning are simple and widely supported. • GPS and Maps APIs can support location awareness during approved outings. • IVR/automated calling services can enable parent approval through phone calls. • Push notifications, SMS, or email can provide timely updates. • Cloud databases can store gate-pass, approval, and entry/exit records securely. • The system can be developed using React, Node.js, and PostgreSQL/MongoDB. • The solution is technically feasible, but privacy, security, service costs, and reliable internet/network connectivity need to be considered.

INSIGHTS EXTRACTED

KEY PAIN POINTS / BARRIERS	KEY GAIN POINTS / EXPECTATIONS
• Faster gate-pass approval • Verified parent permission • QR-based entry/exit • Instant parent notifications • SOS and safety support • Better monitoring and transparency	• Manual and time-consuming process • Approval delays • Difficult parent verification • Manual entry/exit errors • Limited safety monitoring • Privacy and network concerns

INSIGHT STATEMENTS

Write 2–4 evidence-based, non-obvious insights. Each insight should connect an observed behaviour or pattern with an underlying need or motivation.

1	Students want a faster gate-pass process, but they also need **clear approval status and less waiting**.
2	Parents want to approve their child's outing because they need **confidence that the student is safe and the permission is genuine**.
3	Wardens and security staff depend on manual records, showing a need for **quick verification and accurate real-time information**.
4	Students and parents value safety during outings, creating a need for **instant alerts, emergency support, and timely return updates**.

PROBLEM FRAMING

POINT OF VIEW (POV) STATEMENT

[User / stakeholder] needs a way to [need] because [evidence-based insight].

Students need a way to **apply and track gate passes easily** because the manual process can cause delays. Parents need a way to **verify and approve outings quickly** because they want confidence that the permission is genuine. Wardens need a way to **manage approvals and records efficiently** because monitoring students manually is time-consuming. Security staff need a way to **verify student entry and exit quickly** because manual records can cause delays and errors.

HOW MIGHT WE (HMW) STATEMENTS

1. How might we create a secure and simple digital gate-pass system that enables quick parent approval, QR-based entry/exit, timely notifications, and better student safety?

2. How might we **ensure genuine parent approval** before students leave?

3. How might we **reduce the workload of wardens and security staff**?

REVIEW 1 SUBMISSION READINESS

The project problem is stated from the user / stakeholder perspective, not as a technology feature.	**[ ] Yes [ ] No**
Primary research evidence is organized and can be verified by the review panel.	**[ ] Yes [ ] No**
Insights are traceable to research observations, quotations, patterns or artefacts.	**[ ] Yes [ ] No**
The POV and HMW statements remain solution-neutral and allow multiple concepts.	**[ ] Yes [ ] No**
Every team member is prepared to explain their individual contribution during viva.	**[ ] Yes [ ] No**

Team declaration: The submitted field evidence is authentic, ethically collected and accurately represented. Any use of AI tools for transcription, summarisation, translation or visual preparation has been disclosed to the faculty coordinator.

GENERALIZED EVALUATION GUIDE – APPLICABLE TO ALL TECHNOLOGY DOMAINS

Criterion	Max	What the Faculty Evaluates
Research Depth & Authenticity	**5**	Relevant user groups and stakeholders; adequate and diverse primary research; genuine field engagement; context understood; claims not based only on assumptions.
Quality of Evidence Submitted	**3**	Organized, dated and traceable evidence; suitable notes, transcripts, survey/observation records, photos or artefacts; evidence directly supports findings; privacy respected.
Insight Clarity & Relevance	**4**	Findings are synthesized into clear pains, gains, needs and opportunity areas; insights are evidence-based, specific, non-obvious and relevant to the selected application context.
POV & HMW Quality	**3**	POV clearly identifies the user, need and insight. HMW statements are open-ended, actionable, solution-neutral, human-centred and neither too broad nor too narrow.
Understanding of User Research – Viva	**4**	Student can explain methods, users, evidence, findings, limitations, ethical issues and domain context, and can justify how conclusions were reached.
Clarity in Explaining Insights	**3**	Student communicates the research-to-insight-to-problem-framing logic clearly, using specific evidence and appropriate terminology.
Contribution & Involvement	**3**	Student shows verifiable ownership of assigned tasks, participation in field work and synthesis, collaboration, timely documentation and responsible conduct.

Scoring note: Award full marks only when the relevant indicators are demonstrated with convincing evidence. Award proportional marks when evidence is partial, weakly connected, generic or unsupported. Technical sophistication alone should not compensate for weak user research or problem framing in Review 1.

DOMAIN-NEUTRAL INTERPRETATION NOTES

ML / DL / AI	Evaluate the human need, decision context, data availability, representation, privacy, bias and explainability signals. Model accuracy or architecture is not the focus of Review 1.
VR / AR	Evaluate the user setting, device access, interaction needs, safety, motion comfort, spatial constraints, accessibility and the value of immersion or augmentation.
Animation	Evaluate the intended audience, communication or learning need, narrative comprehension, cultural relevance, accessibility, viewing context and media constraints.
Hybrid Projects	Apply the same human-centred criteria across the combined technologies. The team must justify why each technology is relevant to the validated need.
All Domains	Reward evidence-based problem understanding and clear framing. Do not reward a fashionable technology choice when the user need, evidence or opportunity is weak.

REVIEW 1 – MARK DISTRIBUTION (25 MARKS PER STUDENT)

Part A – Team Evaluation (Common Marks) – 15 Marks

Criteria	Max Marks	Marks Awarded	Brief Faculty Observation
Research Depth & Authenticity	5		
Quality of Evidence Submitted	3		
Insight Clarity & Relevance	4		
POV & HMW Quality	3		
Subtotal – Team Score	**15**		

Part B – Individual Evaluation – 10 Marks (Per Student)

Criteria	Max Marks	Student 1 Roll No: _____	Student 2 Roll No: _____	Student 3 Roll No: _____
Understanding of User Research (Viva)	4			
Clarity in Explaining Insights	3			
Contribution & Involvement	3			
Subtotal – Individual Score	**10**			

CONSOLIDATED MARKS

Roll No.	Student Name	Team Score /15	Individual Score /10	Total /25
1.	PradeepKumar R			
2.	Maahesh S			

COORDINATOR COMMENTS AND ACTION POINTS

Strengths / Evidence Accepted / Corrections Required / Next Review Expectations			
[ ] Approved	**[ ] Approved with Revision**	**[ ] Re-review Required**	**Re-review Date: ____ / ____ / 20____**

Signature : Faculty Coordinator (Dr. P. Revathy,Professor,CSD) **Head of the Department**